

HDOC Day-38

Menu-Driven C Program Using Separate Functions

This program performs the following operations on a string using separate functions:

1. Count spaces, digits, and special characters
2. Count the frequency of a given character
3. Toggle the case of each character
4. Exit

Algorithm

1. Start.
2. Declare a string and read it from the user.
3. Display a menu with four choices.
4. If the user selects:
 - a) Choice 1: call countTypes() to count spaces, digits, and special characters.
 - b) Choice 2: read a character and call frequency() to count its occurrences.
 - c) Choice 3: call toggleCase() to change uppercase letters to lowercase and lowercase letters to uppercase.
 - d) Choice 4: terminate the program.
5. Repeat the menu until the user chooses Exit.
6. Stop.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void countTypes(char str[])
{
    int spaces = 0, digits = 0, special = 0;
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ' ')
            spaces++;
        else if (isdigit((unsigned char)str[i]))
            digits++;
        else if (!isalpha((unsigned char)str[i]))
            special++;
    }

    printf("\nSpaces = %d", spaces);
    printf("\nDigits = %d", digits);
    printf("\nSpecial characters = %d\n", special);
}

void frequency(char str[], char ch)
{
    int count = 0, i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
            count++;
    }

    printf("\nFrequency of '%c' = %d\n", ch, count);
}

void toggleCase(char str[])
{
    int i;
```

```

    for (i = 0; str[i] != '\0'; i++)
    {
        if (islower((unsigned char)str[i]))
            str[i] = toupper((unsigned char)str[i]);
        else if (isupper((unsigned char)str[i]))
            str[i] = tolower((unsigned char)str[i]);
    }

    printf("\nString after toggling case: %s\n", str);
}

int main()
{
    char str[200], ch;
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    /* Remove the newline added by fgets */
    str[strcspn(str, "\n")] = '\0';

    do
    {
        printf("\n===== MENU =====");
        printf("\n1. Count spaces, digits and special characters");
        printf("\n2. Count frequency of a character");
        printf("\n3. Toggle case of each character");
        printf("\n4. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                countTypes(str);
                break;

            case 2:
                printf("Enter the character whose frequency is required: ");
                scanf(" %c", &ch);
                frequency(str, ch);
                break;

            case 3:
                toggleCase(str);
                break;

            case 4:
                printf("\nProgram exited successfully.\n");
                break;

            default:
                printf("\nInvalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Sample Test Case

Input string: Hello World 123!

Choice 1 gives: Spaces = 2, Digits = 3, Special characters = 1.

For character 'l', Choice 2 gives: Frequency = 3.

Choice 3 changes the string to: hELLO wORLD 123!

Choice 4 exits the program.

Conclusion

The program successfully implements all the required string operations using separate functions and a menu-driven structure.